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Inventor(s)

Johnson; Graham et al.

SUPPORT ASSEMBLY FOR HOLDING A VIDEOSCOPE ON A ROBOTIC SURGICAL SYSTEM

Abstract

An assembly for holding a videoscope on a robotic surgical device is disclosed, which includes a holder body having a proximal end portion and a distal end portion, a cradle provided on the proximal end portion of the holder body for supporting a videoscope, a first locking mechanism operatively associated with the cradle for temporarily securing the videoscope in the cradle, a connector provided on the distal end portion of the holder body for attaching the holder body to the robotic surgical device, and a second locking mechanism operatively associated with the connector for temporarily securing the holder body to the robotic surgical device.

Inventors: Johnson; Graham (Houston, TX), Park; Yongman (Houston, TX), Cho; Sungwoo (Houston, TX), Kim; Daniel (Houston, TX), Lee; Raymond (Houston, TX), Shin; Dongsuk (Houston, TX)

Applicant: EndoQuest Robotics, Inc. (Houston, TX)

Family ID: 1000007723079

Assignee: EndoQuest Robotics, Inc. (Houston, TX)

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Background/Summary

BACKGROUND OF THE DISCLOSURE

1. Field of the Invention

[0001] The subject disclosure is directed to robotic surgical systems, and more particularly, to a support assembly for temporarily holding a videoscope used with a robotically controlled medical device to enable a surgeon to manually navigate the videoscope to a surgical site under visualization without assistance.

2. Description of Related Art

[0002] In the performance of robotically assisted transluminal and endoluminal surgical procedures performed in or through a natural orifice of a patient, it is necessary to introduce an elongated videoscope camera to the surgical site. Typically, a surgeon will manually navigate the distal tip of the videoscope to the surgical site under visualization, while an assistant holds the proximal portion or housing of the videoscope.

[0003] It would be beneficial however, to enable a surgeon to introduce the videoscope to the surgical site without an assistant holding the videoscope. This would permit the assistant to perform more meaningful tasks during the initial phase of the procedure.

[0004] The subject disclosure provides such a beneficial solution by way of a support assembly for temporarily holding a videoscope to enable a surgeon to manually navigate the distal tip of the videoscope to the surgical site under visualization and without assistance.

SUMMARY OF THE DISCLOSURE

[0005] The subject disclosure is directed to a new and useful assembly for holding a videoscope on a robotic surgical device, which includes an elongated holder body having a proximal end portion and a distal end portion. A cradle is provided on the proximal end portion of the holder body for supporting a videoscope and a first locking mechanism is operatively associated with the cradle for temporarily securing the videoscope in the cradle. A connector is provided on the distal end portion of the holder body for attaching the holder body to the robotic surgical device, and a second locking mechanism is operatively associated with the connector for temporarily securing the holder body to the robotic surgical device.

[0006] The cradle on the proximal end portion of the holder body includes a first engagement feature adapted to receive a complementary engagement feature on the videoscope, and the connector on the distal end portion of the holder body includes a second engagement feature adapted for reception in a complementary engagement feature on the robotic surgical device. In an embodiment of the subject disclosure, the first engagement feature is an engagement channel formed in the cradle of the holder body and the complementary engagement feature on the videoscope is an elongated engagement flange for engaging the engagement channel in the cradle, and the second engagement feature is an elongated engagement flange on the distal end portion of the holder body and the complementary engagement feature on the robotic surgical device is an engagement channel formed in a connective bracket for receiving the engagement flange of the holder body.

[0007] The first locking mechanism includes a cantilevered locking lever and a retention clasp on the cantilevered locking lever for releasably engaging a capture slot in the engagement flange on the videoscope. The second locking mechanism includes a spring-loaded locking slider and a retention tang on the spring-loaded locking slider for releasably engaging a retention groove in the connector of the robotic surgical device. The second locking mechanism further includes a pair of elongated spaced apart retention pins on the spring-loaded locking slider for releasably engaging a pair of corresponding spaced apart retention ports in a proximal end of the robotic surgical device adjacent the connective bracket.

[0008] The assembly further includes a hand rest operatively associated with the distal end portion of the holder body for supporting a user's hand. More particularly, the holder body defines a longitudinal axis and the hand rest extends in an axial direction relative to the distal end portion of the holder body and is mounted for pivotal movement between a first position retracted within an internal cavity of the holder body and a second position extending outwardly from the internal cavity of the holder body. A retention clasp is provided on the hand rest for engaging a retention slot in a side wall of the holder body to retain the hand rest in the second position.

[0009] The subject disclosure is also directed to a method for holding a videoscope on a robotic surgical device, which includes the steps of providing a holder body having a proximal end portion and a distal end portion; temporarily securing the distal end portion of the holder body to the robotic surgical device; and temporarily securing the videoscope on the proximal end portion of the holder body.

[0010] Preferably, the step of temporarily securing the distal end portion of the holder body to the robotic device includes mating an engagement flange on the distal end portion of the holder body with a complementary engagement channel on the robotic surgical device. And the step of temporarily securing the distal end portion of the holder body to the robotic device further includes activating a first locking mechanism that prevents inadvertent disengagement of the engagement flange on the distal end portion of the holder body and the complementary engagement channel on the robotic surgical device.

[0011] Preferably, the step of temporarily securing the videoscope on the proximal end portion of the holder body includes mating an engagement flange on the videoscope with a complementary engagement channel on the proximal end portion of the holder body. And the step of temporarily securing the videoscope on the proximal end portion of the holder body further includes activating a second locking mechanism that prevents inadvertent disengagement of the engagement flange on the videoscope and the complementary engagement channel on the proximal end portion of the holder body.

[0012] The method further includes the step of moving a hand rest from a first position retracted within an internal cavity of the holder body to a second position extending outwardly from the internal cavity of the holder body. And the method also includes the step of securing the hand rest in the second position by engaging a retention clasp on the hand rest in a retention slot formed in the holder body.

[0013] These and other features of the subject disclosure will become more readily apparent to those having ordinary skill in the art to which the subject invention appertains from the detailed description of the preferred embodiments taken in conjunction with the following brief description of the drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0014] So that those skilled in the art will readily understand how to make and use the support assembly of the subject disclosure without undue experimentation, preferred embodiments thereof will be described in detail herein below with reference to the figures wherein:

[0015] FIG. 1 is an illustration of a surgeon utilizing the support assembly while it is attached to a robotic surgical device to introduce a videoscope into a body lumen of a patient;

[0016] FIG. 2 is a side elevational view of the support assembly attached to a robotic surgical device and holding a videoscope with the handle of the support assembly in a deployed position;

[0017] FIG. 3 is a side elevational view of the support assembly detached from a robotic surgical device with the videoscope separated from support assembly;

[0018] FIG. 4 is an illustration showing engagement of a videoscope within the cradle of the

support assembly;

[0019] FIG. **5** is a side elevational view of the support assembly with the side wall broken away to illustrate movement of the handle from a stowed position into a deployed position;

[0020] FIGS. **6** through **8** illustrate the securement of the engagement flange of the videoscope with the cantilevered locking lever of the support assembly;

[0021] FIG. **9** is a perspective view of the support assembly of the subject disclosure;

[0022] FIG. **10** is an enlarged perspective view of the distal end portion of the support assembly with a side wall broken away to illustrate the internal features of the spring-loaded locking mechanism for securing the support assembly to a proximal end of the robotic surgical device;

[0023] FIG. **11** is another perspective view of the support assembly of the subject disclosure;

[0024] FIG. **12** is an illustration showing the complementary engagement features on the distal end of the support assembly and the proximal end of the robotic surgical device;

[0025] FIGS. **13** through **15** illustrate the securement of the engagement flange of the videoscope with the spring loaded locking slider of the support assembly; and

[0026] FIG. **16** illustrates the videoscope and the robot surgical device engaged with the robotic console for use during a robotically assisted surgical procedure after having been separated from the support assembly after the videoscope has been positioned in the body lumen of the patient.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

[0027] Referring now to the drawings, wherein like reference numerals identify similar structural elements and features of the subject disclosure, there is illustrated in FIG. **1** the support assembly **10** of the subject invention, which is temporarily attached to the proximal end of a robotic surgical device **12**. The support assembly **10** is shown holding a videoscope **14** to enable a surgeon to manually navigate the distal end of the videoscope **14** to a surgical site within a body lumen of a patient under visualization, using the manual steering knob **16** of the surgical device **12**, and without assistance from operating room staff.

[0028] A robotic surgical device of the type illustrated in FIG. **1** is disclosed in commonly assigned U.S. Patent Application Publication No. 2023/0210618, the disclosure of which is herein incorporated by reference in its entirety. The robotic surgical device shown in FIG. **1** is designed for use in conjunction with a patient console of a robotic surgical system of the type disclosed in commonly assigned U.S. Patent Application Publication No. 2023/0285098, the disclosure of which is herein incorporated by reference in its entirety.

[0029] As illustrated in FIG. **1**, when the support assembly **10** is temporarily attached to the robotic surgical device **12**, the elongated flexible camera **18** of videoscope **14** is extended through a working channel **17** of the surgical device **12** and into the steerable shaft **19** of the surgical device **12**, which is controlled by the manual steering knob **16** of the surgical device **12**.

[0030] Referring now to FIGS. **2** and **3**, the support assembly **10** of the subject disclosure includes an elongated body **20** having opposed proximal and distal end portions **22** and **24**. The proximal end portion **22** is curved and upturned from the axis of the body **20**. It includes a generally V-shaped cradle **26** for holding the housing **28** of the videoscope **14**. The cradle **26** includes a first side arm **26a** having locking mechanism **30** for temporality securing the videoscope **14** in the cradle **26** and a second side arm **26b** with a frictional retention pad **26c** positioned therein, which is best seen in FIG. **9**.

[0031] Locking mechanism **30** includes a first engagement feature located within the cradle **26** in the form of an elongated engagement channel **32** and a second engagement feature on the housing **28** of videoscope **14** in the form of a complementary rectangular engagement flange **34**, which are best seen in FIG. **4**. The locking mechanism **30** further includes a generally L-shaped cantilevered locking lever **36** fixed at one end by fasteners **38** for temporarily securing the engagement flange **34** with the engagement slot **32**. The structure and function of locking mechanism **30** will be described in more detail below with reference to FIGS. **6** through **8**.

[0032] With continuing reference to FIGS. **2** and **3**, the distal end portion **24** of the body **20** of

support assembly **10** includes a second locking mechanism **40** for temporality securing the body **20** to the proximal end of the robotic surgical device **12**. Locking mechanism **40** includes, among other features, a first engagement feature located on the distal end portion **24** of body **20** in the form of a dovetail-shaped engagement flange **42** and a second engagement feature on the proximal end of the robotic surgical device **12** in the form of a complementary engagement channel **44** formed in a connective bracket **82**, as best seen in FIGS. **11** and **12**. The locking mechanism **40** further includes a spring-loaded locking slider **46** for temporarily securing the engagement flange **44** within the engagement slot **44**, which is best seen in FIGS. **9** and **10**. The structure and function of locking mechanism **40** will be described in more detail below with reference to FIGS. **13** through **15**.

[0033] Referring now to FIG. **5**, the support assembly **10** of the subject disclosure further includes a deployable generally J-shaped hand rest **50** operatively associated with the distal end portion **24** of the holder body **20** for supporting a surgeon's hand during the manual navigation of the distal end of the videoscope **14** to a surgical site. The hand rest **50** extends in an axial direction relative to the distal end portion **24** of the holder body **20** and is mounted for pivotal movement between a first position retracted within an internal cavity **25** of the holder body **20** and a second position extending outwardly and downwardly from the internal cavity **25** of the holder body **20**. In the retracted position, the hand rest **50** is secured from movement through the engagement of retention clasp **54** in a first generally rectangular slot **53** formed in the side wall of the body **20**.

[0034] The hand rest **50** rotates about a pivot pin **55** and is secured in the deployed downward position through the engagement of the retention clasp **54** on the upper end of the hand rest **50** within a second generally rectangular retention slot **57** formed in the side wall of the body **20**. The same hand rest pivot and retention features are shown on the opposite side wall of the support assembly in FIGS. **11** and **12**, and identified as pivot pin **155**, retention clasp **154**, and first and second retention slots **153** and **157**.

[0035] Referring now to FIGS. **6** through **8**, there is illustrated the way the components of locking mechanism **30** function to temporarily secure the videoscope **14** in the cradle **26** of support assembly **10**. Initially, the rectangular engagement flange **34** on the housing **28** of videoscope **14** is aligned with the engagement channel **32** in the cradle **26** of the support assembly **10**. Then, the videoscope **14** is moved in a proximal direction so that the engagement flange **34** slides proximally within the engagement channel **32**. Eventually, the sloped leading edge **34a** of engagement flange **34** meets the sloped front camming face **56a** of a retention clasp **56** that projects upwardly from the top surface of lever **36**, as shown in FIG. **6**.

[0036] As the flange **34** continues to move proximally in channel **32**, the front face **56a** of retention clasp **56** is cammed or otherwise urged in a downward direction until it runs along the undersurface of the engagement flange **34**, as best seen in FIG. **7**. Continued movement of the videoscope **14** in a proximal direction enables the retention clasp **56** to flex back in an upward direction to engage the capture slot **58** formed in the distal end portion of engagement flange **34**, as shown in FIG. **8**. This mechanical engagement releasably secures the position of the videoscope **14** within the cradle **26** of support assembly **10**. Moreover, it prevents the inadvertent disengagement of the videoscope **14** from the cradle **26**. At an appropriate time, the retention clasp **56** can be readily released from the capture slot **58** by pulling the lever **36** downward. Then, the videoscope **14** can be moved distally to disengage flange **34** from channel **32**.

[0037] Referring once again to FIGS. **9** through **12**, as mentioned above, the second locking mechanism **40** for temporality securing the body **20** to the proximal end of the robotic surgical device **12** includes a spring-loaded locking slider **46** for temporarily securing the engagement flange **44** within the engagement slot **44**. Slider **46** is associated with a sliding switch **66** configured for manual movement by a user. A pair of axially retained and parallel arranged compression springs **70a** and **70b** are operatively associated with the proximal surface of the shorter leg **46a** of slider **46** for biasing the slider **46** in a distal direction, which corresponds to a locked position of the

mechanism **40**.

[0038] Locking mechanism **40** further includes an engagement tang **76** which extends proximally from the longer leg **46b** of slide **46**, and through a slot **75** in the proximal end **24** of support structure **10**, for releasably engaging a corresponding retention groove **78** formed within the engagement slot **44** at the proximal end of robotic surgical device **12**, as best seen in FIG. **12**.

[0039] Referring now to FIGS. **13** through **15**, there is illustrated the way the components of locking mechanism **40** function to temporarily secure the distal end portion **24** of the support assembly **10** to the proximal end of the robotic surgical device **12**. Initially, the dovetail shaped engagement flange **42** on the proximal end portion **24** of support assembly **10** is aligned with the engagement channel **44** at the proximal end of surgical device **12** as illustrated in FIG. **12**. Then, the support assembly **10** is moved in a lateral direction so that the engagement flange **42** slides into the engagement channel **44**. Eventually, the leading edge **80** of the connective bracket **82** that forms channel **44** meets the sloped front face **76a** of the engagement tang **76** that projects proximally from slide block **46**, as shown in FIG. **13**.

[0040] Continued lateral movement of the support structure **10** enables the slide block **46** to cammed or otherwise move proximally against the bias of springs **70a**, **70b**, as shown in FIG. **14**. Eventually, the engagement tang **76** will align with retention groove **78** and the slide block **46** will translate distally causing the engagement tang **76** to releasably engage groove **78**. This mechanical engagement releasably secures the support assembly **10** to the proximal end of robotic surgical device **12**. Moreover, it prevents the inadvertent disengagement of the support assembly **10** from the robotic surgical device **12**. At an appropriate time, the surgeon can slide switch **66** in a proximal direction to release the tang **76** from groove **78**, and the engagement flange **42** can be slid laterally out of the slot **44** formed in connective bracket **82**.

[0041] As illustrated in FIG. **16**, when the distal end of the flexible camera **18** has reached the surgical site within the body lumen of the patient, having been navigated there by a surgeon under visualization and without assistance, the distal end portion **24** of support assembly **10** can be released or otherwise disengaged from the proximal end of the robotic surgical device **12**, and the videoscope **14** can be released from the cradle **26** of support assembly **10**. Thereafter, the housing **28** of videoscope **14** can be attached to the robotic surgical system **100** for use during the performance of a surgical procedure, and the support assembly **10** can be stored away for subsequent use.

[0042] While the subject disclosure has been shown and described with reference to preferred embodiments, those skilled in the art will readily appreciate that changes and/or modifications may be made thereto without departing from the scope of the subject disclosure.

Claims

1. A support assembly for holding a videoscope on a robotic surgical device, comprising: a) a holder body having a proximal end portion and a distal end portion; b) a cradle provided on the proximal end portion of the holder body for supporting a videoscope; c) a first locking mechanism operatively associated with the cradle for temporarily securing the videoscope in the cradle; d) a connector provided on the distal end portion of the holder body for attaching the holder body to the robotic surgical device; and e) a second locking mechanism operatively associated with the connector for temporarily securing the holder body to the robotic surgical device.
2. The assembly of claim 1, wherein the cradle on the proximal end portion of the holder body includes a first engagement feature adapted to receive a complementary engagement feature on the videoscope.
3. The assembly of claim 1, wherein the connector on the distal end portion of the holder body includes a second engagement feature adapted for reception in a complementary engagement feature on the robotic surgical device.

- 4.** The assembly of claim 2, wherein the first engagement feature is an engagement channel formed in the cradle of the holder body and the complementary engagement feature on the videoscope is an elongated engagement flange for engaging the engagement channel in the cradle.
- 5.** The assembly of claim 3, wherein the second engagement feature is an elongated engagement flange on the distal end portion of the holder body and the complementary engagement feature on the robotic surgical device is an engagement channel formed in a connective bracket for receiving the engagement flange of the holder body.
- 6.** The assembly of claim 4, wherein the first locking mechanism includes a cantilevered locking lever.
- 7.** The assembly of claim 6, wherein the first locking mechanism further includes a retention clasp on the cantilevered locking lever for releasably engaging a capture slot in the engagement flange on the videoscope.
- 8.** The assembly of claim 5, wherein the second locking mechanism includes a spring-loaded locking slider.
- 9.** The assembly of claim 8, wherein the second locking mechanism further includes a retention tang on the spring-loaded locking slider for releasably engaging a retention groove in the connector of the robotic surgical device.
- 10.** The assembly of claim 8, wherein the second locking mechanism further includes a pair of elongated spaced apart retention pins on the spring-loaded locking slider for releasably engaging a pair of corresponding spaced apart retention ports in a proximal end of the robotic surgical device adjacent the connective bracket.
- 11.** The assembly of claim 1, further comprising a hand rest operatively associated with the distal end portion of the holder body for supporting a user's hand.
- 12.** The assembly of claim 11, wherein the holder body defines a longitudinal axis and the hand rest extends in an axial direction relative to the distal end portion of the holder body and is mounted for pivotal movement between a first position retracted within an internal cavity of the holder body and a second position extending outwardly from the internal cavity of the holder body.
- 13.** The assembly of claim 12, further comprising a retention clasp provided on the hand rest for engaging a retention slot in a side wall of the holder body to retain the hand rest in the second position.
- 14.** A method for holding a videoscope on a robotic surgical device, comprising the steps of: a) providing a holder body having a proximal end portion and a distal end portion; b) temporarily securing the distal end portion of the holder body to the robotic surgical device; and c) temporarily securing the videoscope on the proximal end portion of the holder body.
- 15.** The method of claim 14, wherein the step of temporarily securing the distal end portion of the holder body to the robotic device includes mating an engagement flange on the distal end portion of the holder body with a complementary engagement channel on the robotic surgical device.
- 16.** The method of claim 15, wherein the step of temporarily securing the distal end portion of the holder body to the robotic device further includes activating a first locking mechanism that prevents inadvertent disengagement of the engagement flange on the distal end portion of the holder body and the complementary engagement channel on the robotic surgical device.
- 17.** The method of claim 14, wherein the step of temporarily securing the videoscope on the proximal end portion of the holder body includes mating an engagement flange on the videoscope with a complementary engagement channel on the proximal end portion of the holder body.
- 18.** The method of claim 17, wherein the step of temporarily securing the videoscope on the proximal end portion of the holder body further includes activating a second locking mechanism that prevents inadvertent disengagement of the engagement flange on the videoscope and the complementary engagement channel on the proximal end portion of the holder body.
- 19.** The method of claim 14, further comprising the step of moving a hand rest from a first position retracted within an internal cavity of the holder body to a second position extending outwardly

from the internal cavity of the holder body.

20. The method of claim 19, further comprising the step of securing the hand rest in the second position by engaging a retention clasp on the hand rest in a retention slot formed in the holder body.
